Road accidents in Italy: where to invest?

I looked for the Italian towns with the most accidents: in total, per inhabitant and per km².

168,557

accidents per year

3,075

deaths per year

28.6

accidents per 10,000 inhabitants

Italy, average 2022-2024. Data: ISTAT (accidents) and SITUAS (population and area).

What I did

1. Downloaded the ISTAT data with the API and saved it on my PC
2. Got population and area from SITUAS with web scraping
3. Cleaned the data and joined the two tables
4. Calculated accidents per 10,000 inhabitants and per km²
5. Grouped the towns with clustering (KMeans)
6. ANOVA test and a simple forecast (linear regression)
7. Built a dashboard in Power BI

Problems I solved

- API limit: 5 calls per minute, so I saved the data once

- 2024: towns with 0 accidents were missing, I counted them as 0

- Very small towns have extreme rates: I used towns with 10,000+ inhabitants

Tools

Tool	Used for
Python + pandas	download, cleaning, new columns
requests	ISTAT API and SITUAS scraping
matplotlib, seaborn	charts
scikit-learn	clustering (KMeans)
scipy, statsmodels	ANOVA and regression
Power BI	dashboard
Git, GitHub	save and share the project

What I found



The drop in 2020 is because of the COVID lockdown.

- Most accidents per inhabitant: tourist towns and big cities like Bergamo, Florence, Genoa

- Most accidents per km²: Milan, Florence, Turin, Naples

- Centre and North have more accidents than South and Islands (ANOVA)

- 3 groups of towns: high risk per inhabitant, dense cities, fewer but more serious accidents

- Forecast: a small decrease, about 167,000 accidents in 2025

Where to invest

1. High risk per inhabitant

326 towns, mostly Centre and North

Prevention and road safety checks

2. Dense cities

87 towns (Milan, Naples, Turin...)

Better traffic management

3. Fewer but more serious accidents

789 towns

Speed control on extra-urban roads